INDIVIDUAL CONTRIBUTION REPORT

Student Club Event Ticketing & Budget Portal

Software Engineering — Labs 1, 2 and 3

Team member	Deeksha G
SRN	PES1UG24AM078
Lab 3 responsibility	Component modelling and interface specification
Jira allocation	SCRUM-21–SCRUM-24; 16 points

1. Purpose and reporting basis

This report consolidates the member's documented responsibilities across the project. It explains the recorded work, how it was approached, relevant Jira items, and planned implementation and validation. Completed analysis/modelling deliverables are distinguished from future software implementation.

2. Whole-project context

- **Purpose:** A centralized portal for student-club event proposals, budget decisions, QR-ticket generation, ticket booking/access and registration monitoring.
- **Actors:** Club Lead, Campus Admin and Student.
- **Lab 1 foundation:** Requirements engineering and use-case modelling: actors, functional requirements, non-functional requirements, priorities and use-case flows. The available records do not provide a reliable individual authorship log for Lab 1.
- **Lab 2 outputs:** Eight Jira Epics, sixteen User Stories, priorities, story-point estimates, team ownership, two sprint plans, board/status tracking, burndown, timeline and reflection.
- **Lab 3 outputs:** Architecture comparison and decision; Layered Architecture selected; five components; six interfaces; UML component diagram; written architectural justification.
- **Core functional scope:** Submit proposals and budgets; review and approve/reject; generate unique QR tickets after approval; purchase tickets after successful payment; view/download tickets; monitor registrations and availability.
- **Quality requirements:** Role-based authentication/authorization; at least 95% of requests should respond within 3 seconds under expected load.
- **Status boundary:** The evidence reviewed documents analysis, modelling, planning and Jira tracking. It does not establish that a complete production application, payment integration, QR-validation workflow or performance-tested deployment exists.

3. Individual responsibility and outputs

Lab 3 responsibility: Component modelling and interface specification. Documented output: Identification of five major components, six service interfaces, and component dependencies.

Jira allocation: Epics 5–6; SCRUM-21–SCRUM-24; 16 estimated story points out of 71 across the backlog.

4. What was completed or prepared

- Structured the student-facing journey into browsing approved events, purchasing a ticket, viewing the ticket and downloading it.
- Owned Event Ticket Booking and QR Ticket Access stories in Jira, totalling 16 estimated points.
- Contributed to component modelling: Authentication & Authorization, Event & Proposal Management, Ticket Management, Payment Service, and Registration & Event Management.
- The interface specification records AuthenticationAPI, ProposalAPI, PaymentAPI, RegistrationAPI, EventDataAPI and TicketDataAPI.
- The model establishes payment confirmation as a prerequisite for issuing a usable ticket to the student.

5. How the work was achieved

1. Followed the student journey from selecting an approved event through purchase and ticket retrieval.
2. Kept browsing, purchasing, viewing and downloading as separate user goals so each could be prioritized, estimated and tracked.
3. Estimated purchase higher than browse/view/download because it spans payment, ticket issuance and registration/availability updates.
4. Named the interfaces and identified provider/consumer components rather than leaving component communication implicit.
5. Mapped PaymentAPI to Payment Service and Ticket Management, and ticket/registration interfaces to the components responsible for records and availability.

6. Jira work items

Issue	Story	Points	Sprint	Purpose
SCRUM-21	Browse Approved Events	3	Sprint 2	Browse approved events
SCRUM-22	Purchase Event Ticket	8	Sprint 2	Purchase ticket after successful payment
SCRUM-23	View Purchased QR Ticket	3	Sprint 2	View purchased QR ticket
SCRUM-24	Download QR Ticket	2	Sprint 2	Download purchased QR

				ticket

The Jira records reviewed for this report show these stories in Done status. A Done status reflects the tracked work-item state; it is not, by itself, proof that a working application feature has been coded and tested.

7. Lab-by-lab contribution narrative

Lab 1 — Requirements Engineering and Use-Case Modelling

The shared requirements foundation identifies Club Lead, Campus Admin and Student, and covers proposal/budget submission, administrative review, ticket generation after approval, ticket purchase/access, registration monitoring, role-based access and response-time expectations. These requirements informed the Jira backlog and Lab 3 architecture. Available records do not assign exclusive Lab 1 authorship to individual members, so this is described as a shared baseline.

Lab 2 — Agile Backlog Creation and Sprint Simulation

The Jira backlog contains eight Epics and sixteen stories, with priorities, estimates and team ownership. The member owns SCRUM-21–SCRUM-24 (16 points). Two sprints were planned and tracked with board, burndown, timeline and summary evidence. Current Jira records show the member's listed stories as Done.

Lab 3 — Component Modelling and Architecture Selection

The team compared Layered, Microservices and Client–Server and selected Layered Architecture. Five components were documented: Authentication & Authorization; Event & Proposal Management; Ticket Management; Payment Service; Registration & Event Management. Six interfaces were specified: AuthenticationAPI, ProposalAPI, PaymentAPI, RegistrationAPI, EventDataAPI and TicketDataAPI.

8. Planned work and next steps

6. Implement a student catalogue that exposes only approved events.
7. Integrate a payment gateway or clearly defined payment simulation with success, failure and cancellation states.
8. Issue tickets only after payment confirmation; restrict ticket retrieval to the purchaser.
9. Test failed payments, repeat purchase attempts, unavailable capacity and unauthorized ticket access.
10. Validate QR payloads and consistency between ticket records, event capacity and registrations.

9. Dependencies and validation criteria

Important dependencies are approval before ticket generation, successful payment before ticket issuance, consistency between capacity and registrations, and authorization on protected operations. Test normal and failure paths including rejected proposals, failed payments, full events, unauthorized access, duplicate purchases and slow responses.

The performance target—at least 95% of requests within 3 seconds under expected load—is a requirement to verify through measurement, not a result to claim without test evidence.

10. Evidence boundary and status

Available records confirm booking/access story ownership and component/interface modelling, but do not prove that payment integration or a student ticketing interface has been implemented.

Documented at analysis/planning/modelling level: requirements and use-case foundation, Jira Epics/stories, estimates and sprint planning, architecture comparison, component model, interfaces, UML diagram, architectural justification and reports. Implementation/verification still needs evidence unless separately completed: executable workflows, payment integration, QR validation, automated tests, concurrency checks, security tests and measured performance tests.

11. Overall contribution summary

Deeksha G is documented as responsible for component modelling and interface specification in Lab 3 and for SCRUM-21–SCRUM-24 (16 story points) in Jira. The work connects project requirements to an implementation plan. The next phase is to build and integrate the components through defined interfaces and collect test evidence.

Prepared from the team's Lab 2/Lab 3 reports and Jira issue records reviewed for this report.